Project Initialization and Planning Phase

Date	27 October 2026
Team ID	EN25CS3011046
Project Title	Formula 1 Race Intelligence Dashboard
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build an interactive Formula 1 Race Intelligence Dashboard that turns 6,464 race outcomes from 17 seasons into clear visual insights on driver, team, circuit and tyre performance, and to make it available to anyone through a website.
Scope	Loading and cleaning the F1 race dataset; building a Tableau dashboard and story (driver leaderboard, points trajectory, team ranking, championship analysis, DNF reasons, tyre distribution, podiums and country participation); publishing it to Tableau Public; and embedding it in a Flask web page. Live race telemetry and prediction models are outside the scope.
Problem Statement	
Description	Race analysts, fans and journalists must search through thousands of race records and static tables to compare drivers and teams across seasons. Without one interactive view, finding patterns is slow and conclusions are hard to verify.
Impact	Solving this lets users find insights in minutes instead of hours, compare performance across seasons, understand reliability and tyre strategy, and share findings through a simple link. It supports better-informed analysis, reporting and discussion.

Proposed Solution	
Approach	Collect and load the dataset, then clean and join the tables in Tableau. Define KPIs and calculated fields, build the charts and dashboard with filters, and create a story. Publish to Tableau Public, embed the views in a Flask page using HTML and the Tableau Embedding API, test the interactivity, and document the work on GitHub.
Key Features	Interactive filters by season, driver and team; driver leaderboard, points trajectory, team ranking and championship analysis; DNF reliability and tyre compound views; podium and country participation views; a Tableau story from overview to championship success; free, shareable web access through Flask.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop or PC (e.g., Intel Core i5, 4 cores); no GPU needed
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	About 1 GB free disk space for Tableau, the dataset and project files
Software		
Frameworks	Python frameworks	Flask (Python 3)
Libraries	Additional libraries	Flask, Jinja2 (installed with Flask), Tableau Embedding API (JavaScript)
Development Environment	IDE, version control	VS Code, Tableau Desktop, Git and GitHub

Data		
Data	Source, size, format	Public Formula 1 race results dataset: about 6,464 race outcomes across 17 seasons, in CSV/Excel format